

Genome-wide association analysis of yield reduction under water deficit stress reveals key loci associated with drought tolerance in soybean

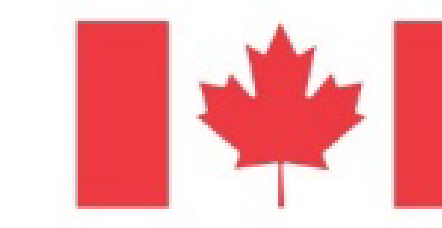
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Introduction: Soybean is a globally important crop contributing to food security and environmental sustainability thanks to its symbiotic nitrogen fixation [1]. Water deficit stress and drought are increasing with climate change, and can have a significant impact on yield in almost all growing regions even when total precipitation is adequate but inconsistently delivered [2]. Selection of drought tolerant genotypes for crop breeding is a critical challenge relying on high quality phenotypic assessment as well as genomics techniques such as genome-wide association study (GWAS).

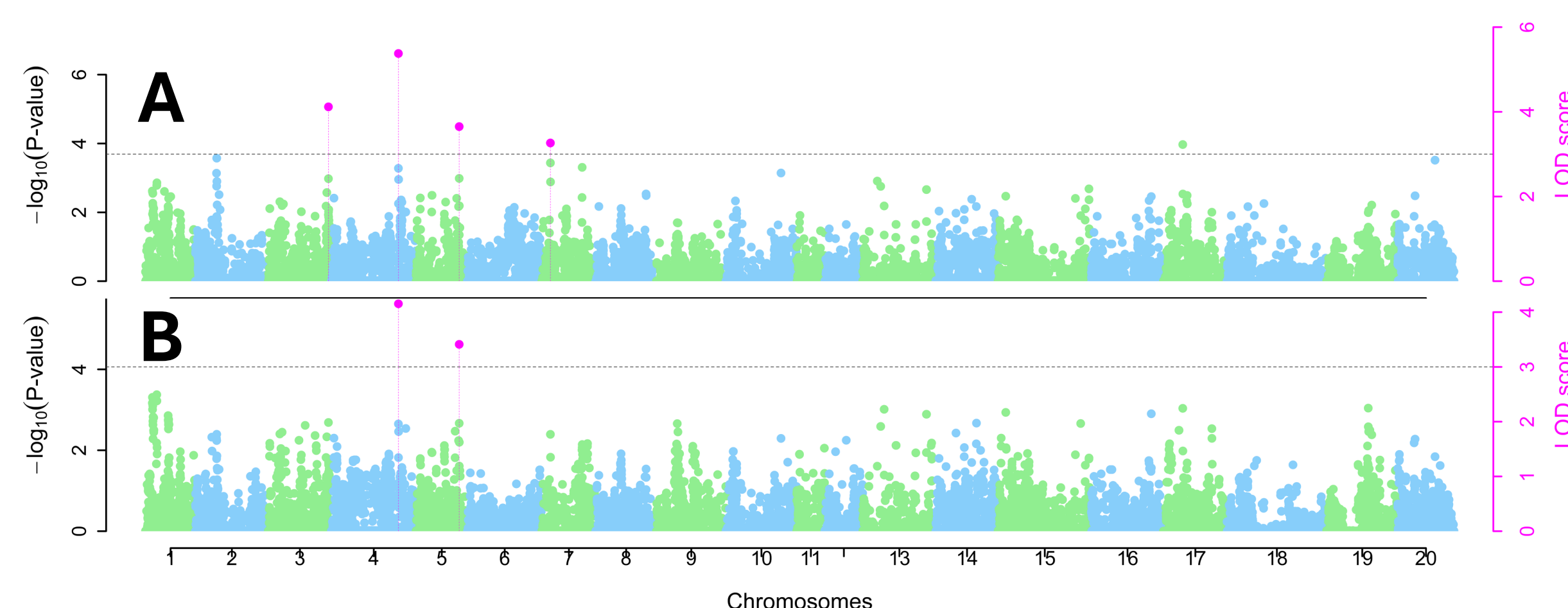


Figure 1: Manhattan plots of GWAS mrMLM analysis for A) $\Delta YLD\%$ and B) $\Delta SDm2\%$.

Methods: A panel of Canadian soybean genotypes ($n=137$) was grown in innovative side by side irrigated and non-irrigated yield trial experiments over four years. Phenotypes representing tolerance to water deficit stress were generated by comparing genotype performance between the two treatment conditions (yield and yield components: $\Delta YLD\%$, $\Delta SDm2\%$, $\Delta SDWT$, $\Delta SDWT\%$; seed composition: ΔP , $\Delta P\%$, ΔO , $\Delta O\%$). ANOVA was carried out in R [3] and multi-locus GWAS was carried out using 19139 SNPs and the mrMLM v4.02 method [4,5]. Haplotype blocks were delineated [6], and allele substitution effect was calculated [7]. Post-GWAS analysis was carried out using the Accuracy approach [8], the AccuCalc package [9], and the Genomic Variation Explorer [10].

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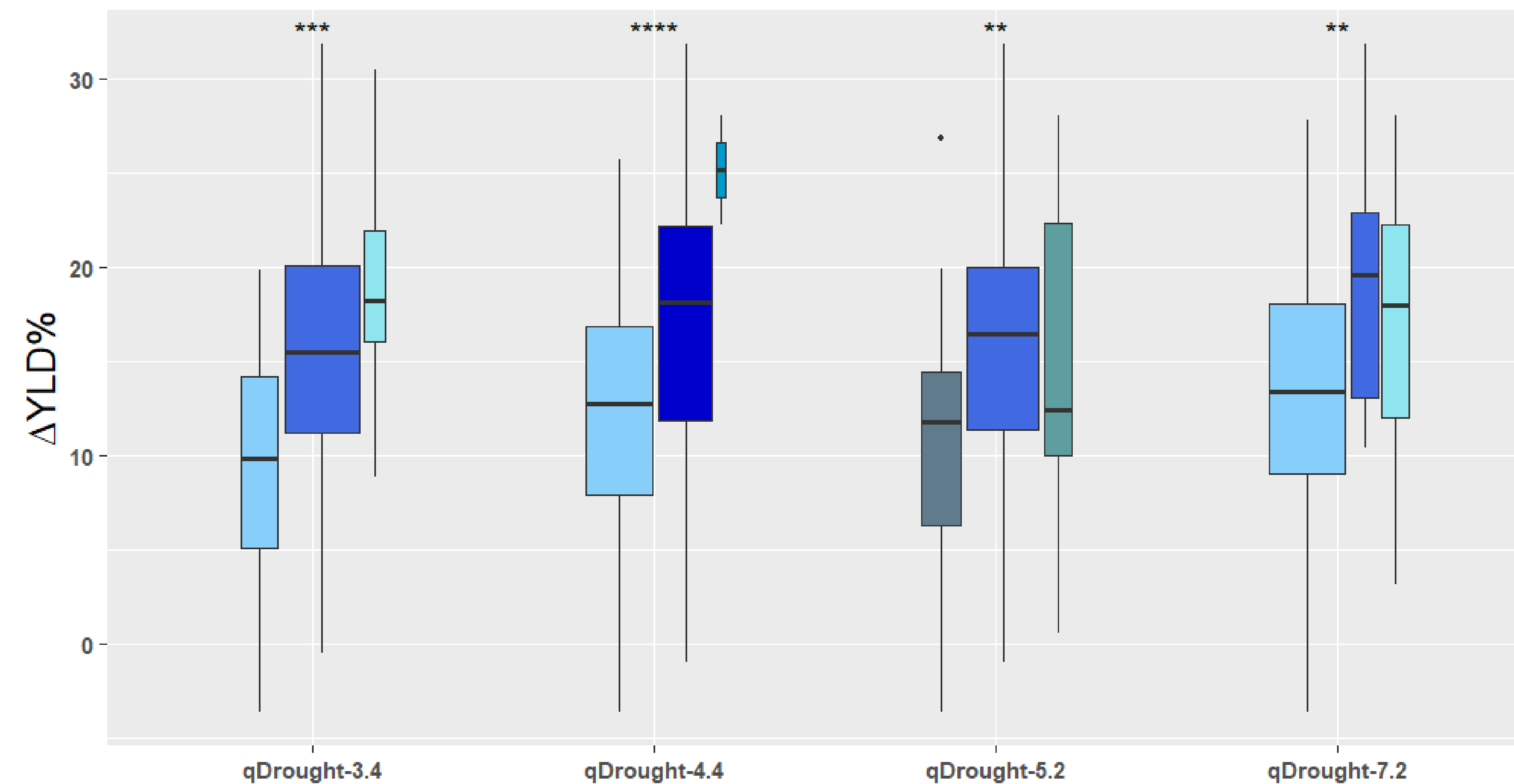


Figure 2: ASE for $\Delta YLD\%$ detected in short season Canadian soybean tested under irrigated and non-irrigated treatments.

Results: Irrigation of soybean yield trial plots delivered between 29% and 42% additional water availability compared to the non-irrigated treatment over the four years studied. However, yield loss due to water deficit stress was between 0% and 30%, indicating that genotypic response is not uniform. The effect of irrigation on phenotypic response was significant for yield, seed weight, seeds per m^2 , and oil, but not significant for seed protein content. Association analysis revealed associations on chromosomes 3, 4, 5, 6, 7, 8, 14, 16, and 20 for traits derived from comparison between treatments. Four QTL for $\Delta YLD\%$ (*qDrought-3.4*, *qDrought-4.4*, *qDrought-5.2*, and *qDrought-7.2*) have ASE between 4.64% and 6.32% yield loss between allelic classes. Post-GWAS analysis for $\Delta YLD\%$ QTL revealed important variants, one of which, *Glyma.03G25*****, has high impact variants in the promoter region that affect the binding of transcription factors.

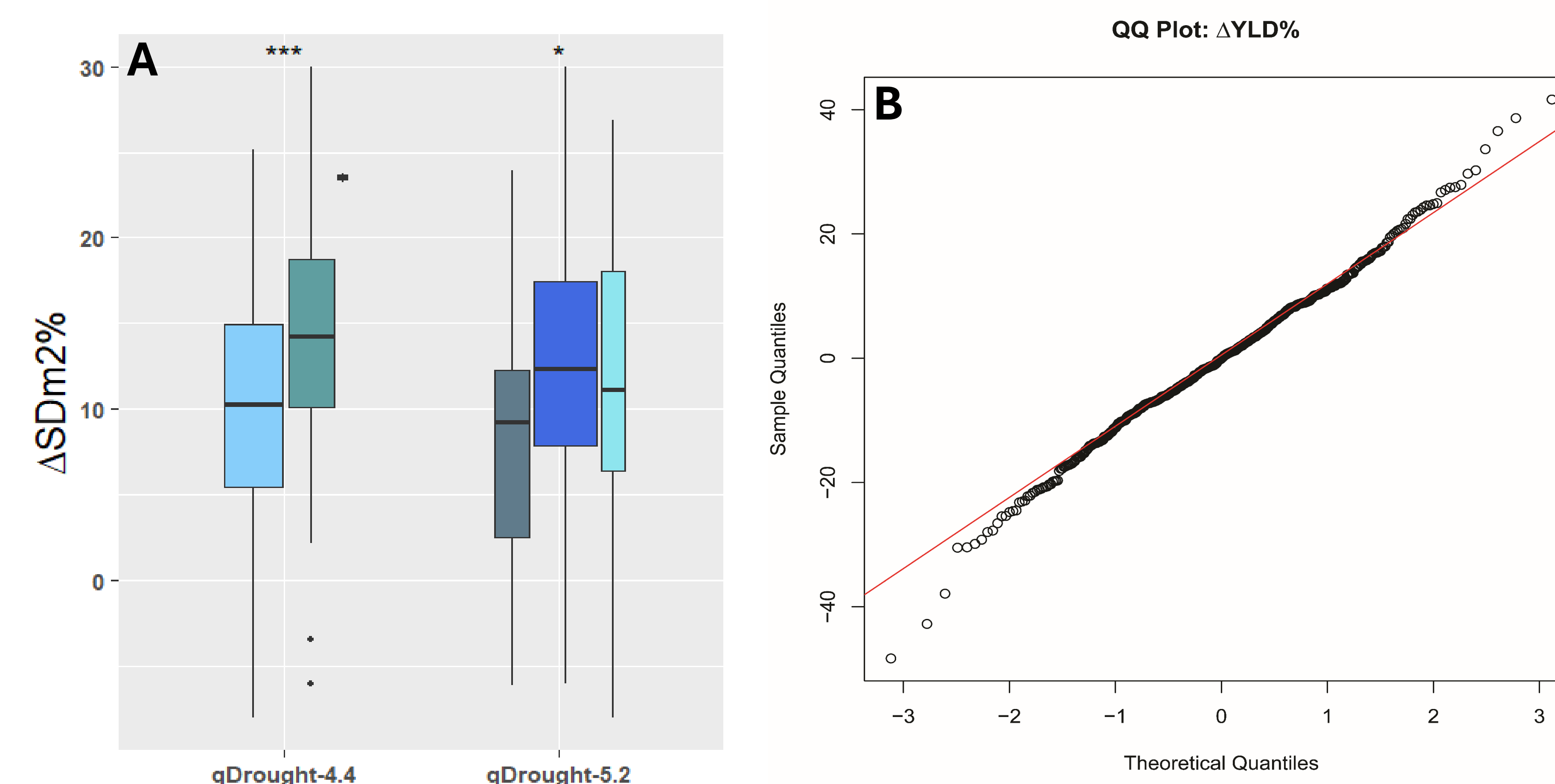


Figure 3: A) ASE for $\Delta SDm2\%$ detected in short season Canadian soybean tested under irrigated and non-irrigated treatments. B) Phenotypic QQ plot of $\Delta YLD\%$ showing normality of the data.

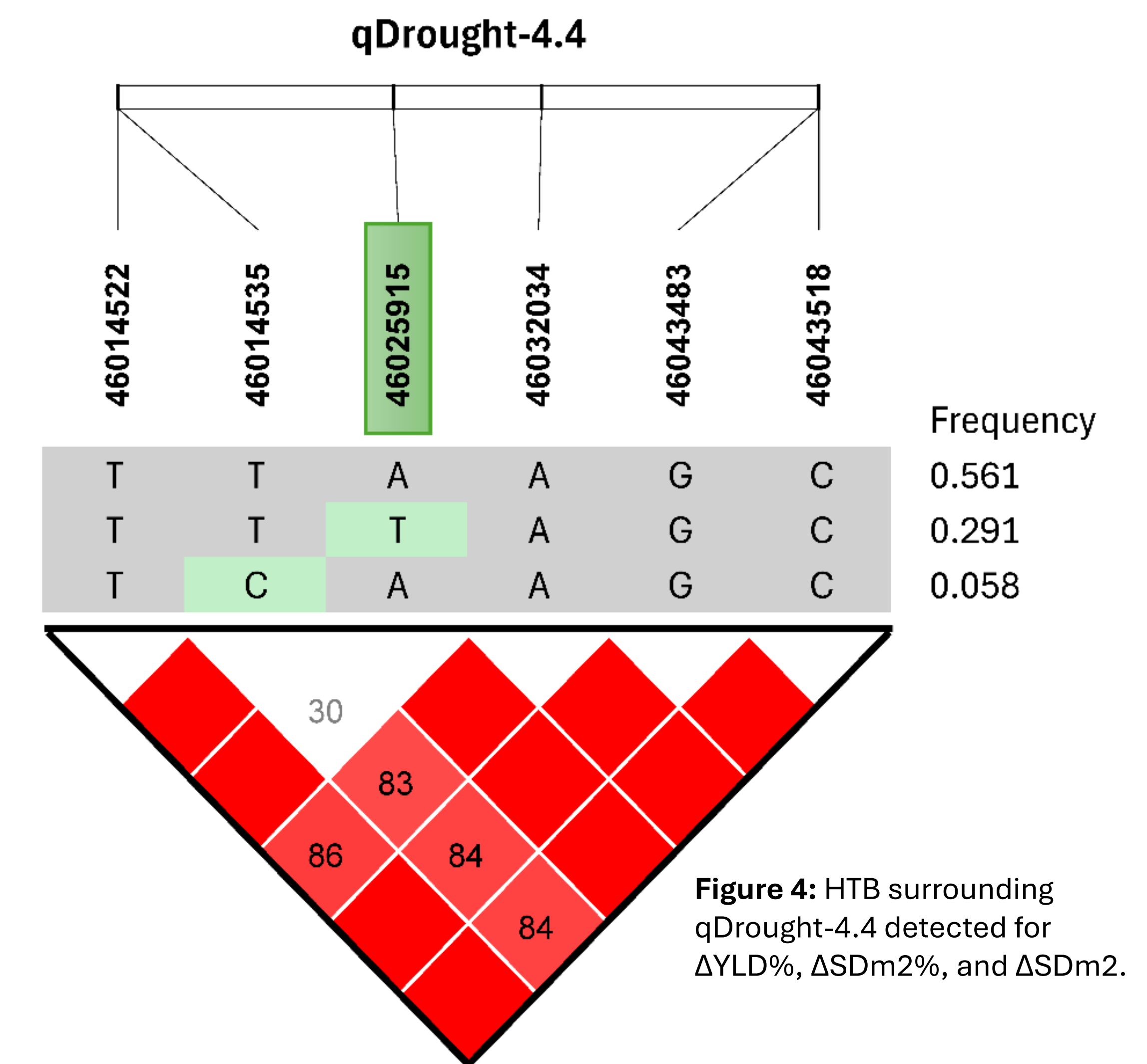


Figure 4: HTB surrounding *qDrought-4.4* detected for $\Delta YLD\%$, $\Delta SDm2\%$, and $\Delta SDm2\%$.

Discussion and Conclusion: Climate change is predicted to increase the incidence of drought and cause between 7% and 23% in crop losses across regions and species requiring costly changes in crop inputs to mitigate [11]. Spring planted soybean in temperate regions are susceptible to water deficit stress as their yield production is largely controlled by moisture availability during critical pod filling periods. Reducing yield penalty due to water deficit stress by breeding for enhanced climate resilience will be a key factor in ensuring the ongoing success of Canadian soybean production. Pyramiding of QTL such as the four detected in this study for tolerance to water deficit stress in desirable genetic backgrounds could prove foundational for the next generation of crop breeding. Understanding of molecular pathways, such as changes in transcription factor binding for UDP-glycosyltransferase genes (*Glyma.03G25*****), and how they can be manipulated to combat climate change, represents a major avenue for ongoing study in crop science and beyond.

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References:

